

Question Paper

Exam Date & Time: 24-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Semantic Web [IT_4412]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A software company wants to maintain a digital contact list of its employees to share information easily across different departments and applications. Each contact should include the employee's name, phone number, and email address in a structured format that can be exchanged between systems. Apply your understanding of XML structure to develop an XML document that represents the company's contact list, ensuring that each contact entry contains the required details - name, phone number, and email - in a well-formed and organized manner. (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) RDF data can be represented using different syntaxes such as RDF/XML, Turtle, and JSON-LD, but all convey the same underlying data model. Apply your understanding of RDF representations to explain how these different syntaxes maintain the same semantic meaning. Develop a brief theoretical explanation highlighting how RDF's triple-based structure ensures data consistency across multiple formats. (5)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) An e-commerce company manages a large product catalog that includes information about products, suppliers, prices, and categories. All this information is represented and stored using RDF to enable semantic querying and data integration across multiple vendors. The company wants to analyse supplier performance and pricing trends using SPARQL queries. Analyse how SPARQL queries, using various filters, pattern matching, and construct clauses, can be designed to extract all suppliers offering products priced below ₹1000. (5)
- A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) A healthcare startup is planning to design and implement an ontology for managing and integrating patient health records. Examine how adopting different ontology development approaches could influence the accuracy, consistency, and interoperability of the resulting healthcare data model. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) A job portal system aims to build an intelligent platform that can match candidates with suitable job roles based on their skills. To enable smarter data representation and reasoning, the development team decides to use OWL2 to design an ontology that is compatible with RDF/RDFS standards. Apply OWL2 to create an ontology ensuring RDF/RDFS compatibility. (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) A semantic reasoning module in a chatbot must infer facts quickly. Choose an OWL2 profile that (5)

suits this need and explain how description logic supports reasoning under constraints.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) In an online certification system, once a badge is awarded, it cannot be revoked. Develop a monotonic SWRL rule to model this and explain its significance in maintaining trust. (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) In an energy grid, power supply rules change dynamically depending on demand. Develop RuleML-based non-monotonic rules to capture this scenario. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5) A travel agency wants to create a system that combines flight details, hotel options, and local attractions to help customers plan their trips easily. To build this system efficiently, the agency decides to use existing travel ontologies instead of starting from scratch. Apply your understanding of ontology reuse and extension to develop a domain ontology model that integrates flight, hotel, and attraction data, ensuring that the information is interoperable and reusable across different travel services. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Two universities maintain their own academic ontologies - one uses "Course" and "Instructor," while the other uses "Module" and "Teacher" to represent similar concepts. To enable data sharing and integration between their systems, they need to perform ontology mapping. Apply your understanding of ontology mapping to develop a method that aligns equivalent classes and properties (e.g., Course ↔ Module, Instructor ↔ Teacher) so that data from both universities can be integrated and queried seamlessly across systems. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4412:Semantic Web

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	20
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4412:Semantic Web

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	8	40(80%)
4	Analysis	2	10(20%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)

Total	10	50 (100%)
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